Rudder Trim Position Transmitter – General data

1. General

A potentiometer is mounted on the starboard side of the rudder internally at approximately station 571.978 and sends a signal of the rudder trim position through the DAU and onto the Flight Data Recorder (if installed) and the rudder trim position is indicated on the system screen.

2. Removal/Installation

Equipment and Materials:

- Corrosion preventative compound MIL–C–16173 Grade 2 or equivalent.

A. Remove the Rudder Trim Indicator Potentiometer

1) Ensure the electrical system is OFF (refer to 24-00-00, Electrical Power – General Data).

2) Pull the FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit safety clip.

3) Disconnect the electrical connection at the terminal block.

4) Loosen the nut and screw that hold the lever assembly to the potentiometer shaft mounted in mounting bracket. Remove lever assembly.

5) Remove the (2) bolts, nuts and washers that hold the potentiometer mounting bracket. Remove the potentiometer and the mounting bracket assembly.

6) Remove the safety wire and the (3) screws and clamps which secure the potentiometer onto the mounting bracket.

7) Remove potentiometer.

B. Install the Rudder Trim Indicator Potentiometer

Note Coat all bolt shanks with corrosion preventative compound MIL–C–16173 Grade 2 or equivalent before installation.

1) Ensure the electrical system is OFF (refer to 24-00-00, Electrical Power – General Data).

2) Ensure the FLIGHT CONTROLS INDICATION (A14) circuit breaker is pulled and fitted with safety clip.

3) Position the potentiometer on to the mounting bracket. Secure with the (3) clamps and screws. Safety wire screws.

4) Position the mounting bracket on channel and secure in place with bolt, (2) washers and nut in two places.

5) Install the lever assembly on the potentiometer shaft ensuring the mechanism can move freely through the full range of motion and there is a minimum gap of 0.04 inch between the lever assembly and the potentiometers housing.

6) Secure with a washer, nut and screw.

7) Install the clevis pin, (2) washers cotter pin from the end of the lever assembly to the link halves.

8) Install the clevis pin, (2) washers cotter pin from the end of the link halves to the end of the push rod assembly.

9) Ensure that the mechanism can move freely through the full range of motion.

10) Adjust the rudder trim indicator potentiometer. (Refer to Para 3.A).

11) Reconnect the electrical connection at terminal block.

12) Remove the safety clip and replace circuit breaker for the FLIGHT CONTROLS INDICATION (A14) circuit breaker.

13) Do a function test of the rudder trim tab indicator potentiometer per Para 3.B.

3. Adjustment/Test

A. Adjustment Rudder Trim Tab Indicator Potentiometer

Equipment and Material:

- DC Voltmeter

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Pull the FLIGHT CONTROLS INDICATION (A14) circuit breaker and fit safety clip.

3) Move rudder to neutral and lock in place (refer to 27-20-00, Rudder Control System – General Data). Set rudder trim tab to the neutral position.

4) Disconnect terminal 2 of the potentiometer electrical connection at the terminal block and connect voltmeter positive to the lead from terminal 2 while attaching voltmeter negative to aircraft ground.

CAUTION **ENSURE THE TERMINAL 2 LEAD DOES NOT COME INTO CONTACT WITH AN ELECTRICAL GROUND.**

5) Loosen screw and nut on lever assembly at the potentiometer shaft enough so that the shaft turns freely.

6) Reset the FLIGHT CONTROLS INDICATION (A14) circuit breaker.

7) Rotate potentiometer shaft until DC Voltage is precisely 50% of total aircraft voltage (approximately 14 volts), pull FLIGHT CONTROLS INDICATION (A14) circuit breaker.

8) Disconnect voltmeter from potentiometer. Reconnect pushrod and lever assembly to potentiometer shaft.

Note Do not rotate potentiometer shaft during this process.

9) Tighten screw and nut on lever assembly leaving a gap of 0.02 to 0.05 of and an inch between lever assembly and potentiometer housing.

10) Reconnect the potentiometer electrical connections at the terminal block.

11) Remove safety clip and replace FLIGHT CONTROLS INDICATION (A14) circuit breaker.

12) Remove rudder control lock.

13) Do a function test per Para 3.B.

14) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

Note For Flight Data Recorder input checks, refer to 31-31-00, Flight Data Recorder (FDR) – General Data.

For Data Acquisition Unit input checks, refer to 31-44-00, Data Acquisition Unit (DAU) (Mod 6/2049) – General Data.

B. Function Test Rudder Trim Tab Indicator Potentiometer

1) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

2) Wait several minutes for the aircraft electrical and avionics systems to initialize.

3) Deflect rudder trim to the full nose left position (Reference 25 degrees) and note yaw trim display indication correlates and does not become invalid (indicated by an amber ‘X’).

4) Deflect rudder trim to the neutral position (Reference 0 degrees) and note yaw trim display indication correlates.

5) Deflect rudder trim to the full nose right position (Reference 15 degrees) and note yaw trim display indication correlates and does not become invalid (indicated by an amber ‘X’).

6) If the rudder trim display does not match the rudder trim surface, repeat the adjustment steps in Para 3.A.

7) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

This Page Intentionally Left Blank